

**PROFIT, LOSS & DISCOUNT**

Cost Price (CP) – The goods are bought at this price.

Selling Price (SP) – The goods are sold at this price.

Profit (When  $SP > CP$ ) =  $SP - CP$  and Loss (When  $CP > SP$ ) =  $CP - SP$

$$\text{Profit Percent} = \frac{P}{CP} \times 100 \text{ and Loss Percent} = \frac{L}{CP} \times 100$$

Marked Price (MP) – This is the price which is listed / marked on the article.

Discount (D) – The difference of MP and SP is known as discount.

$$\text{Discount Percent} = \frac{D}{MP} \times 100 \text{ and Mark-up Percent} = \frac{\text{Markup value}}{CP} \times 100$$

**EXERCISE**

1. Find the CP of an article sold at Rs.2700 at a profit of 12.5%.

- (a) 2400 (b) 1800  
(c) 2750 (d) 3000

2. Find the SP of an article sold at a profit of  $16\frac{2}{3}\%$  if its CP is Rs.720.

- (a) 750 (b) 840  
(c) 780 (d) 810

3. A man sells his goods at 25% loss. If he bought it at Rs.240 then, find the SP of the article.

- (a) 120 (b) 100  
(c) 180 (d) 150

4. A man sells his goods at 25% loss on SP. Find his actual Profit /loss percent.

- (a) 25% (b) 20%  
(c) 16.66% (d) 14.28%

5. A man sells his goods at 16.66% profit on SP. Find his actual Profit /loss percent.

- (a) 25% (b) 20%  
(c) 16.66% (d) 14.28%

6. A man sells an item for Rs.530 and makes a loss of 20%. Find the Cost Price of that item.

- (a) Rs.662.5 (b) Rs.573.5  
(c) Rs.628.5 (d) Rs.549.5

7. If CP of 15 articles is equal to SP of 12 articles. Find Profit/Loss percent.

- (a) 30% (b) 25%  
(c) 20% (d) 40%

8. Undertaker sells an article at a profit of 20% and another at a loss of 20%. If selling price of both the articles is same then, find his profit/loss percent.

- (a) No profit No loss (b) 4% loss  
(c) 4% profit (d) 2% profit

9. John Cena sells an article at a profit of 20% and another at a loss of 20%. If cost price of both the articles is same then, find his profit/loss percent.

- (a) No profit, No loss (b) 4% loss  
(c) 4% profit (d) 2% loss

10. Two mobiles are purchased for Rs.11000. One is sold at 40% profit and the other at 20% loss. If both the mobiles are sold at the same price then, find the cost price of the mobile sold at a loss.

- (a) Rs.7000 (b) Rs.3600  
(c) Rs.5600 (d) Rs.6000

11. After selling 72 articles a man loses SP of 9 articles. Find his loss percent.

- (a) 12.5% (b)  $11\frac{1}{9}\%$   
(c)  $9\frac{1}{11}\%$  (d) 10%

12. After selling 72 articles a man earns a profit of SP of 9 articles. Find his profit percent.

- (a) 7.14% (b) 11.11%  
(d) 14.28% (d) 12.5%

13. After selling 72 articles a man earns a profit of CP of 9 articles. Find his profit percent.

- (a) 12.5% (b)  $11\frac{1}{9}\%$   
(c)  $9\frac{1}{11}\%$  (d) 10%

**14.** The profit earned when an article is sold for Rs.2000 is 3 times the loss incurred when it is sold for Rs.800. Find at what price he must sold the same so that he earns 20% profit.

- (a) 1100 (b) 1250  
(c) 1280 (d) 1320

**15.** A person sells an article at 8% loss. Had he sold it for Rs.450 more, he would gain 7%. Find his initial Cost Price.

- (a) 1500 (b) 2000  
(c) 2400 (d) 3000

**16.** A person sells his goods at 10% profit. Had he purchased it for 20% less and sold it for Rs.40 more, he would gain 40%. Find his initial Cost Price.

- (a) 2000 (b) 1800  
(c) 1000 (d) 3000

**17.** A person purchases 15 articles at Rs.20 and sold 12 articles at Rs.30. Find his loss/Profit percentage.

- (a) 62.5% Loss (b) 62.5% Profit  
(c) 87.5% Loss (d) 87.5% Profit

**18.** A person buys some articles at the rate of 6 for 5 rupees and sells it at the rate of 5 for 6 rupees. State his profit%.

- (a) 36% (b) 40%  
(c) 44% (d) 48%

**19.** A person purchases some articles at Rs.13579. He sold  $\frac{2}{3}$ rd of them at 5% Profit. At what profit percent he must sell the remaining so that he earns 7% overall.

- (a) 9% (b) 11%  
(c) 10% (d) 12%

**20.** A merchant sold an article at Rs.56. If profit percent is numerically equal to its cost price then, find cost price of the article.

- (a) 30 (b) 70  
(c) 40 (d) 60

**21.** A dishonest shopkeeper promise to sell his goods at CP but he uses 20% less weight. Find his profit percentage.

- (a) 25% (b) 18%  
(c) 20% (d) 30%

**22.** A shopkeeper promise to sell his goods at 10% profit but he uses 20% less weight. Find his profit percentage.

- (a) 20% (b) 25%  
(c) 37.5% (d) 45.45%

**23.** A shopkeeper promise to sell his goods at x% profit but uses 20% less weight thus gaining 37.5% overall. Find the value of x.

- (a) 10% (b) 12.5%  
(c) 14.28% (d) 15%

### DISCOUNT

**24.** Find the single equivalent discount of two successive discounts of 20% and 15%.

- (a) 35% (b) 38%  
(c) 32% (d) None of these

**25.** A shopkeeper marks his goods 32% more than the cost price. If he allows a discount of 25% then, his gain/loss percent is?

- (a) 2% (b) 1%  
(c) 3% (d) None of these

**26.** A shopkeeper gives 1 item free on every purchase of 4 item, find the discount % that a customer gets.

- (a) 25% (b) 20%  
(c) 15% (d) 10%

**27.** A shop is offering a scheme in which the customer gets '4 free' articles for every '5 articles purchased'. Find the discount percent offered by the shopkeeper.

- (a) 36.36% (b) 20%  
(c) 44.44% (d) 80%

**28.** After giving 20 percent discount, a man still gains 10% profit. Find ratio of cost price and marked price.

- (a) 5:4 (b) 10:11  
(c) 2:3 (d) 8:11

**30.** A shopkeeper allows 20% discount on marked price and earns 30% profit. If he gets Rs.180 as profit then, find the amount of discount offered.

- (a) 195 (b) 180  
(c) 210 (d) 165